

# FH fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes FH gene fusions from the msk\_impact\_50k\_2026 study, covering 16 fusion events. 1/16 fusions (6.2%) were in-frame. 2/16 fusions (12.5%) retained the Lyase domain. Domain retention was tested with Fisher's exact test ( $p=0.133333$ , not statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for FH.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 16 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was DGKH, observed in 5 events.

### Domain retention

Lyase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 1/1 (100.0%) of in-frame fusions retained the domain,  $p=0.133333$  (not statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.287129.

### Domain disruption

Domain-disruption analysis was skipped: FH has no disruption\_required\_domains configured; domain-disruption analysis was skipped.

### Cutpoint detection

Cutpoint detection scanned 15 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 45 aa (exon 2; permutation-corrected  $p=0.0689311$ , not statistically significant at  $\alpha=0.05$ ). This is -13 aa from the nearest configured domain boundary at 58 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=2$ ) and "not\_retained" ( $n=13$ ). For Frame\_status, retained: 1/1 (100.0%, 95% CI 20.7%-100.0%); not\_retained: 0/5 (0.0%, 95% CI 0.0%-43.4%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 12 vs 29.3077,  $p=0.0682804$  (not statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score reported: domain\_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption\_required\_domains not configured for this gene).

### Exon retention

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## **Expression association**

Expression association reported: No mRNA expression data was available for FH in this cohort; expression-association analysis was skipped.

## **Joint partner**

Joint partner reported: FH has no gene\_pair configured; joint-partner analysis was skipped.

## **Mutation cooccurrence**

Mutation cooccurrence reported: FH has no mutual\_exclusivity\_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

## **Window detection**

Window detection produced results for this run (20 row(s) in window\_scan, 11 row(s) in top\_windows); see the accompanying table.

# Tables

## composite\_score tables

composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                  | Composite_score     | Rank |
|--------------|-------------|------------------|------------------------|-------------------------|--------------------------|----------------------------|------------------------------------------------------------------------|---------------------|------|
| RYR2         | 1           | 0.0625           | 0.05419233758836865    | None                    | 1.0                      | 0.3860334247970314         | confidence_certainty, cutpoint_proximity, domain_retention, recurrence | 0.27612678359599413 | 1    |
| unknown      | 1           | 0.0625           | 0.05419233758836865    | None                    | 1.0                      | 0.3860334247970314         | confidence_certainty, cutpoint_proximity, domain_retention, recurrence | 0.27612678359599413 | 2    |
| PDE1C        | 1           | 0.0625           | 0.05419233758836865    | None                    | 0.9976851851851852       | 0.3860334247970314         | confidence_certainty, cutpoint_proximity, domain_retention, recurrence | 0.27569275581821634 | 3    |
| DGKH         | 5           | 0.3125           | 0.05419233758836865    | None                    | 0.20416666666666666      | 0.3860334247970314         | confidence_certainty, cutpoint_proximity, domain_retention, recurrence | 0.22065803359599415 | 4    |
| NOTCH1       | 1           | 0.0625           | 0.05419233758836865    | None                    | 0.6736111111111112       | 0.3860334247970314         | confidence_certainty, cutpoint_proximity, domain_retention, recurrence | 0.2149288692932747  | 5    |
| RGS7         | 2           | 0.125            | 0.05419233758836865    | None                    | 0.3912037037037037       | 0.3860334247970314         | confidence_certainty, cutpoint_proximity, domain_retention, recurrence | 0.18541497804043855 | 6    |
| FMN2         | 1           | 0.0625           | 0.05419233758836865    | None                    | 0.3055555555555556       | 0.3860334247970314         | confidence_certainty, cutpoint_proximity, domain_retention, recurrence | 0.14591845026266081 | 7    |
| ARID4B       | 1           | 0.0625           | 0.05419233758836865    | None                    | 0.24768518518518523      | 0.3860334247970314         | confidence_certainty, cutpoint_proximity, domain_retention, recurrence | 0.13506775581821637 | 8    |
| TRMT1L       | 1           | 0.0625           | 0.05419233758836865    | None                    | None                     | 0.3860334247970314         | confidence_certainty, domain_retention, recurrence                     | 0.10907911827199278 | 9    |
| MAFTRR       | 1           | 0.0625           | 0.05419233758836865    | None                    | 0.02314814814814814      | 0.3860334247970314         | confidence_certainty, cutpoint_proximity, domain_retention, recurrence | 0.0929670613737719  | 10   |
| SLC35F3      | 1           | 0.0625           | 0.05419233758836865    | None                    | 0.0                      | 0.3860334247970314         | confidence_certainty, cutpoint_proximity, domain_retention, recurrence | 0.08862678359599413 | 11   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio | p_value             | neg_log10_p_value  |
|----------|------------------------|------------------|------------------------|------------------|------------|---------------------|--------------------|
| 44       | 0                      | 2                | 1                      | 12               | 0.0        | 1.0                 | -0.0               |
| 45       | 2                      | 0                | 1                      | 12               | None       | 0.02857142857142857 | 1.5440680443502757 |
| 186      | 2                      | 0                | 2                      | 11               | None       | 0.05714285714285714 | 1.2430380486862944 |
| 246      | 2                      | 0                | 3                      | 10               | None       | 0.09523809523809523 | 1.021189299069938  |
| 345      | 2                      | 0                | 4                      | 9                | None       | 0.14285714285714285 | 0.8450980400142568 |
| 370      | 2                      | 0                | 10                     | 3                | None       | 1.0                 | -0.0               |
| 464      | 2                      | 0                | 11                     | 2                | None       | 1.0                 | -0.0               |
| 467      | 2                      | 0                | 12                     | 1                | None       | 1.0                 | -0.0               |

## domain\_retention tables

### frame\_domain\_contingency\_table

|   |    |
|---|----|
| 1 | 1  |
| 0 | 13 |

### domain\_retention\_descriptives

| Domain_key      | Domain_name | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|-----------------|-------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| auto_key_domain | Lyase       | 15                      | 2                    | 9               | 4                | 0.45713623725671915                             |

## exon\_retention tables

### Exon\_retention

| Exon | Retained_event_count | Total_event_count | Retained_fraction |
|------|----------------------|-------------------|-------------------|
| 2    | 8                    | 16                | 0.5               |

## frequency tables

### Partner\_gene\_counts

| Partner_gene | Event_count |
|--------------|-------------|
| ARID4B       | 1           |
| DGKH         | 5           |

| Partner_gene | Event_count |
|--------------|-------------|
| FMN2         | 1           |
| MAFTRR       | 1           |
| NOTCH1       | 1           |
| PDE1C        | 1           |
| RGS7         | 2           |
| RYR2         | 1           |
| SLC35F3      | 1           |
| TRMT1L       | 1           |
| unknown      | 1           |

## window\_detection tables

### window\_scan

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio | p_value             | neg_log10_p_value  | event_ids_inside                                                                                                                                             |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|------------|---------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -14      | 186    | 200      | 4               | 11               | 2                 | 0                  | None       | 0.05714285714285714 | 1.2430380486862944 | EVT-P-0000532-T02-IM5-25,<br>EVT-P-0015961-T03-IM6-28,<br>EVT-P-0040956-T01-IM6-22,<br>EVT-P-0048392-T01-IM6-1                                               |
| 44       | 244    | 200      | 4               | 11               | 2                 | 0                  | None       | 0.05714285714285714 | 1.2430380486862944 | EVT-P-0000532-T02-IM5-25,<br>EVT-P-0015961-T03-IM6-28,<br>EVT-P-0040956-T01-IM6-22,<br>EVT-P-0048392-T01-IM6-1                                               |
| 170      | 370    | 200      | 9               | 6                | 0                 | 2                  | 0.0        | 0.14285714285714285 | 0.8450980400142568 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0032745-T01-IM6-26,<br>EVT-P-0040956-T01-IM6-22,<br>... (9 total) |
| 186      | 386    | 200      | 9               | 6                | 0                 | 2                  | 0.0        | 0.14285714285714285 | 0.8450980400142568 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0032745-T01-IM6-26,<br>EVT-P-0040956-T01-IM6-22,<br>... (9 total) |
| 246      | 446    | 200      | 8               | 7                | 0                 | 2                  | 0.0        | 0.2                 | 0.6989700043360187 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0032745-T01-IM6-26,<br>EVT-P-0059293-T01-IM7-10,<br>... (8 total) |
| 264      | 464    | 200      | 8               | 7                | 0                 | 2                  | 0.0        | 0.2                 | 0.6989700043360187 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0059435-T01-IM7-11,<br>... (8 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio | p_value             | neg_log10_p_value  | event_ids_inside                                                                                                                                              |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|------------|---------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 267      | 467    | 200      | 9               | 6                | 0                 | 2                  | 0.0        | 0.14285714285714285 | 0.8450980400142568 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0059435-T01-IM7-11,<br>... (9 total)  |
| 270      | 370    | 100      | 7               | 8                | 0                 | 2                  | 0.0        | 0.4666666666666667  | 0.3309932190414244 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0062340-T01-IM7-5,<br>... (7 total)   |
| 277      | 477    | 200      | 10              | 5                | 0                 | 2                  | 0.0        | 0.09523809523809523 | 1.021189299069938  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0051358-T01-IM6-29,<br>EVT-P-0059293-T01-IM7-10,<br>... (10 total) |
| 320      | 370    | 50       | 7               | 8                | 0                 | 2                  | 0.0        | 0.4666666666666667  | 0.3309932190414244 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0062340-T01-IM7-5,<br>... (7 total)   |
| 345      | 370    | 25       | 7               | 8                | 0                 | 2                  | 0.0        | 0.4666666666666667  | 0.3309932190414244 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0062340-T01-IM7-5,<br>... (7 total)   |
| 345      | 395    | 50       | 7               | 8                | 0                 | 2                  | 0.0        | 0.4666666666666667  | 0.3309932190414244 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0062340-T01-IM7-5,<br>... (7 total)   |
| 345      | 445    | 100      | 7               | 8                | 0                 | 2                  | 0.0        | 0.4666666666666667  | 0.3309932190414244 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0062340-T01-IM7-5,<br>... (7 total)   |
| 345      | 545    | 200      | 10              | 5                | 0                 | 2                  | 0.0        | 0.09523809523809523 | 1.021189299069938  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0051358-T01-IM6-29,<br>EVT-P-0059293-T01-IM7-10,<br>... (10 total) |
| 364      | 464    | 100      | 7               | 8                | 0                 | 2                  | 0.0        | 0.4666666666666667  | 0.3309932190414244 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0059435-T01-IM7-11,<br>... (7 total)  |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio | p_value             | neg_log10_p_value   | event_ids_inside                                                                                                                                             |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|------------|---------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 367      | 467    | 100      | 8               | 7                | 0                 | 2                  | 0.0        | 0.2                 | 0.6989700043360187  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0059435-T01-IM7-11,<br>... (8 total) |
| 370      | 395    | 25       | 6               | 9                | 0                 | 2                  | 0.0        | 0.4857142857142857  | 0.31361912297200173 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0062340-T01-IM7-5,<br>... (6 total)  |
| 370      | 420    | 50       | 6               | 9                | 0                 | 2                  | 0.0        | 0.4857142857142857  | 0.31361912297200173 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0062340-T01-IM7-5,<br>... (6 total)  |
| 370      | 470    | 100      | 8               | 7                | 0                 | 2                  | 0.0        | 0.2                 | 0.6989700043360187  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0059435-T01-IM7-11,<br>... (8 total) |
| 370      | 570    | 200      | 9               | 6                | 0                 | 2                  | 0.0        | 0.14285714285714285 | 0.8450980400142568  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0051358-T01-IM6-29,<br>EVT-P-0059293-T01-IM7-10,<br>... (9 total) |

### top\_windows

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio | p_value             | neg_log10_p_value  | event_ids_inside                                                                                                                                              |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|------------|---------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -14      | 186    | 200      | 4               | 11               | 2                 | 0                  | None       | 0.05714285714285714 | 1.2430380486862944 | EVT-P-0000532-T02-IM5-25,<br>EVT-P-0015961-T03-IM6-28,<br>EVT-P-0040956-T01-IM6-22,<br>EVT-P-0048392-T01-IM6-1                                                |
| 277      | 477    | 200      | 10              | 5                | 0                 | 2                  | 0.0        | 0.09523809523809523 | 1.021189299069938  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0051358-T01-IM6-29,<br>EVT-P-0059293-T01-IM7-10,<br>... (10 total) |
| 170      | 370    | 200      | 9               | 6                | 0                 | 2                  | 0.0        | 0.14285714285714285 | 0.8450980400142568 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0032745-T01-IM6-26,<br>EVT-P-0040956-T01-IM6-22,<br>... (9 total)  |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio | p_value             | neg_log10_p_value   | event_ids_inside                                                                                                                                             |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|------------|---------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 267      | 467    | 200      | 9               | 6                | 0                 | 2                  | 0.0        | 0.14285714285714285 | 0.8450980400142568  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0059435-T01-IM7-11,<br>... (9 total) |
| 370      | 570    | 200      | 9               | 6                | 0                 | 2                  | 0.0        | 0.14285714285714285 | 0.8450980400142568  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0051358-T01-IM6-29,<br>EVT-P-0059293-T01-IM7-10,<br>... (9 total) |
| 367      | 467    | 100      | 8               | 7                | 0                 | 2                  | 0.0        | 0.2                 | 0.6989700043360187  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0059435-T01-IM7-11,<br>... (8 total) |
| 246      | 446    | 200      | 8               | 7                | 0                 | 2                  | 0.0        | 0.2                 | 0.6989700043360187  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0032745-T01-IM6-26,<br>EVT-P-0059293-T01-IM7-10,<br>... (8 total) |
| 264      | 464    | 200      | 8               | 7                | 0                 | 2                  | 0.0        | 0.2                 | 0.6989700043360187  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0059435-T01-IM7-11,<br>... (8 total) |
| 345      | 370    | 25       | 7               | 8                | 0                 | 2                  | 0.0        | 0.4666666666666667  | 0.3309932190414244  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0062340-T01-IM7-5,<br>... (7 total)  |
| 364      | 464    | 100      | 7               | 8                | 0                 | 2                  | 0.0        | 0.4666666666666667  | 0.3309932190414244  | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0059435-T01-IM7-11,<br>... (7 total) |
| 370      | 395    | 25       | 6               | 9                | 0                 | 2                  | 0.0        | 0.4857142857142857  | 0.31361912297200173 | EVT-P-0004918-T02-IM6-27,<br>EVT-P-0018108-T01-IM6-7,<br>EVT-P-0018108-T02-IM6-8,<br>EVT-P-0059293-T01-IM7-10,<br>EVT-P-0062340-T01-IM7-5,<br>... (6 total)  |

**Per-event results (results.tsv)**



| event_id                        | sample_id                 | patient_id | tumor_type                 | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                                                            | source_site_effect_on_frame |
|---------------------------------|---------------------------|------------|----------------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| EVT-P-004<br>8392-T01-I<br>M6-1 | P-004<br>8392-T01-I<br>M6 |            | Breast Cancer              | BRC A         |             | unknown      | unknown      | three_prime | 241680684          | 2               | 45                          | True                   | 45                                            | True                                | exact match                         | retained      | 1.0                      | False               | Lyase            |              |                   | FH (NM_00143) rearrangement: c.133-68:FGChr1: g.240716629inv   Transcript Fusion {MIR1273E:FGH}                                   | NA                          |
| EVT-P-006<br>2340-T01-I<br>M7-5 | P-006<br>2340-T01-I<br>M7 |            | Non-Small Cell Lung Cancer | LUAD          | ARID4B-FH   | ARID4B       | out-of-frame | three_prime | 241665896          | 8               | 370                         | True                   | 370                                           | True                                | exact match                         | disrupted     | 0.060240963855421686     | True                |                  |              | Lyase             | ARID4B (NM_016374) - FH (NM_00143) rearrangement: c.1841+5245:ARID4B_c.1109-26:FGHdup   Protein Fusion: out of frame {ARID4B:FGH} | NA                          |

| event_id                    | sample_id             | patient_id | tumor_type                 | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail                                                                                                                                                                                | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                                                 | source_site2_effect_on_frame |
|-----------------------------|-----------------------|------------|----------------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>8108-T01-IM6-7 | P-001<br>8108-T01-IM6 |            | Non-Small Cell Lung Cancer | LUSC          | DGKH-FH     | DGKH         | unknown      | five_prime  | 241667339          | 7               | 370                         | True                   | 369                                           | True                                | hybrid junction condition: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | disrupted     | 0.9427710843373494       | True                |                  |              | Lyase             | FH (NM_001430143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667340::chr13:g.42830153)   Transcript Fusion {FH:DGKH} | NA                           |

| event_id                    | sample_id             | patient_id | tumor_type                 | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail                                                                                                                                                                                | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                       | source_site2_effect_on_frame |
|-----------------------------|-----------------------|------------|----------------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|----------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>8108-T02-IM6-8 | P-001<br>8108-T02-IM6 |            | Non-Small Cell Lung Cancer | LUSC          | DGKH-FH     | DGKH         | unknown      | five_prime  | 241667340          | 7               | 370                         | True                   | 369                                           | True                                | hybrid junction condition: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | disrupted     | 0.9427710843373494       | True                |                  |              | Lyase             | FH (NM_001430143)(chr14:11,141,166,734,000-11,141,166,734,000)   Transcript Fusion {FH:DGKH} | NA                           |

| event_id                    | sample_id             | patient_id | tumor_type     | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail                                                                                                                                                                            | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                                                | source_site2_effect_on_frame |
|-----------------------------|-----------------------|------------|----------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-006<br>4901-T01-IM7-9 | P-006<br>4901-T01-IM7 |            | Ovarian Cancer | HGSC          | DGKH-FH     | DGKH         | unknown      | five_prime  | 241667338          | 7               | 370                         | True                   | 369                                           | True                                | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | disrupted     | 0.9427710843373494       | True                |                  |              | Lyase             | FH (NM_0010143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667338::chr13:g.42830159)   Transcript Fusion {FH:D GKH} | NA                           |

| event_id                         | sample_id                 | patient_id | tumor_type                | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail                                                                                                                                                                            | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                                                      | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|---------------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-005<br>9293-T01-I<br>M7-10 | P-005<br>9293-T01-I<br>M7 |            | Cancer of Unknown Primary | CUP           | DGKH-FH     | DGKH         | unknown      | five_prime  | 241667341          | 7               | 370                         | True                   | 369                                           | True                                | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | disrupted     | 0.9427710843373494       | True                |                  |              | Lyase             | FH (NM_0010143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667341::chr13:g.42830159)   Protein Fusion: mid-exon {FH:DGKH} | NA                           |

| event_id                   | sample_id           | patient_id | tumor_type             | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                                             | source_site2_effect_on_frame |
|----------------------------|---------------------|------------|------------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-005-9435-T01-I-M7-11 | P-005-9435-T01-I-M7 |            | Esophagogastric Cancer | SPDAC         | DGKH-FH     | DGKH         | unknown      | three_prime | 241661272          | 10              | 464                         | True                   | 464                                           | True                                | exact match                         | lost          | 0.0                      | False               |                  | Lyase        |                   | FH (NM_00143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241661272::chr13:g.42830525)   Transcript Fusion {DGKH:FH} | NA                           |

| event_id                         | sample_id                 | patient_id | tumor_type           | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail                                                                                                                                                                                | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                               | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|----------------------------------------------------------------------|------------------------------|
| EVT-P-006<br>6922-T02-I<br>M7-20 | P-006<br>6922-T02-I<br>M7 |            | Hepatobiliary Cancer | GBC           | FMN2-FH     | FMN2         | unknown      | five_prime  | 241667416          | 7               | 345                         | False                  | 344                                           | True                                | hybrid junction condition: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | disrupted     | 0.8674698795180723       | True                |                  |              | Lyase             | FH (NM_0010143) rearrangement: chr10:g.240469632del Antisense Fusion | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type        | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                                                              | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0060877-T01-IM7-21 | P-0060877-T01-IM7 |            | Colorectal Cancer | COAD          | MAFTRR-FH   | MAFTRR       | unknown      | three_prime | 241661260          | 10              | 467                         | False                  | 467                                           | True                                | exact match                         | lost          | 0.0                      | False               |                  | Lyase        |                   | FH (NM_0010143) rearrangement: t(1;16)(q43;q23.2)(chr1:g.241661260::chr16:g.79754421)   Protein Fusion: mid-exon {LOC102467146: FH} | NA                           |



| event_id                         | sample_id                 | patient_id | tumor_type  | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                                                                               | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-004<br>0956-T01-I<br>M6-22 | P-004<br>0956-T01-I<br>M6 |            | Anal Cancer | ANSC          | NOTCH1-FH   | NOTCH1       | out-of-frame | three_prime | 241672198          | 5               | 186                         | True                   | 186                                           | True                                | exact match                         | disrupted     | 0.6144578313253012       | True                |                  |              | Lyase             | NOTCH1 (NM_017617) - FH (NM_000143) rearrangement: t(1;9)(q43;q34.3)(chr1:g.241672198::chr9:g.139417940)   Protein Fusion: out of frame {NOTCH1: FH} | NA                           |

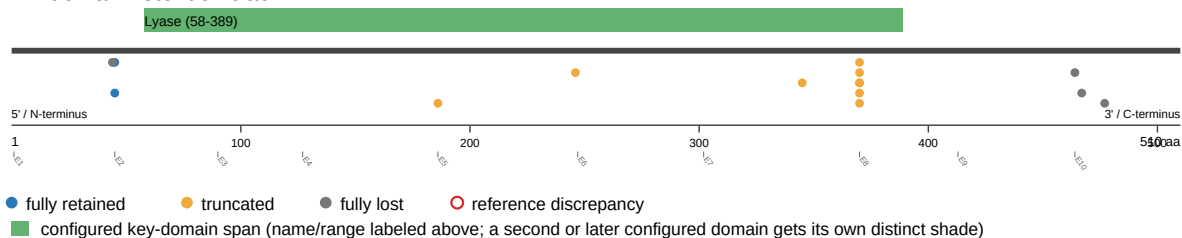
| event_id                 | sample_id         | patient_id | tumor_type    | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                 | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0000532-T02-IM5-25 | P-0000532-T02-IM5 |            | Breast Cancer | MDLC          | PDE1C-FH    | PDE1C        | out-of-frame | five_prime  | 241682866          | 1               | 44                          | True                   | 44                                            | True                                | exact match                         | lost          | 0.0                      | False               |                  | Lyase        |                   | FH (NM_001143) - PDE1C (NM_001191058) Rearrangement : t(1;7)(1q43;7p14.3)(chr1:g.241682866::chr7:g.31926696)   Protein fusion: out of frame (FH-PDE1C) | NA                           |

| event_id                         | sample_id                 | patient_id | tumor_type    | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                                                    | source_site_effect_on_frame |
|----------------------------------|---------------------------|------------|---------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| EVT-P-003<br>2745-T01-I<br>M6-26 | P-003<br>2745-T01-I<br>M6 |            | Mesothelioma  | PLSM<br>ESO   | RGS7-FH     | RGS7         | out-of-frame | five_prime  | 241671722          | 5               | 246                         | True                   | 246                                           | True                                | exact match                         | disrupted     | 0.5692771084337349       | True                |                  |              | Lyase             | FH (NM_00143) - RGS7 (NM_002924) rearrangement: c.738+181:FH_c.386-8492:RGS7del   Protein Fusion: out of frame {FH:RGS7}  | NA                          |
| EVT-P-000<br>4918-T02-I<br>M6-27 | P-000<br>4918-T02-I<br>M6 |            | Breast Cancer | IDC           | RGS7-FH     | RGS7         | out-of-frame | three_prime | 241666009          | 8               | 370                         | True                   | 370                                           | True                                | exact match                         | disrupted     | 0.060240963855421686     | True                |                  |              | Lyase             | RGS7 (NM_002924) - FH (NM_00143) rearrangement: c.79-95591:RGS7_c.1109-139:FHdup   Protein Fusion: out of frame {RGS7:FH} | NA                          |

| event_id                  | sample_id          | patient_id | tumor_type                 | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains | disrupted_domains | source_annotation_text                                                                                                            | source_site2_effect_on_frame |
|---------------------------|--------------------|------------|----------------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001-5961-T03-IM6-28 | P-001-5961-T03-IM6 |            | Non-Small Cell Lung Cancer | LUAS          | RYR2-FH     | RYR2         | in-frame     | three_prime | 241680785          | 2               | 45                          | True                   | 45                                            | True                                | exact match                         | retained      | 1.0                      | False               | Lyase            |              |                   | RYR2 (NM_001035) - FH (NM_00143) rearrangement: c. 48+5657: RYR2_c.133-169: FHinv   Protein Fusion: in frame {RYR2: FH}           | NA                           |
| EVT-P-005-1358-T01-IM6-29 | P-005-1358-T01-IM6 |            | Colorectal Cancer          | READ          | SLC35F3-FH  | SLC35F3      | unknown      | three_prime | 241661231          | 10              | 477                         | False                  | 477                                           | True                                | exact match                         | lost          | 0.0                      | False               |                  | Lyase        |                   | SLC35F3 (NM_173508) - FH (NM_00143) rearrangement: c. 283+157829: SLC35F3_c.1430: F Hinv   Protein Fusion: mid-exon {SLC35F3: FH} | NA                           |

## domain retention outliers

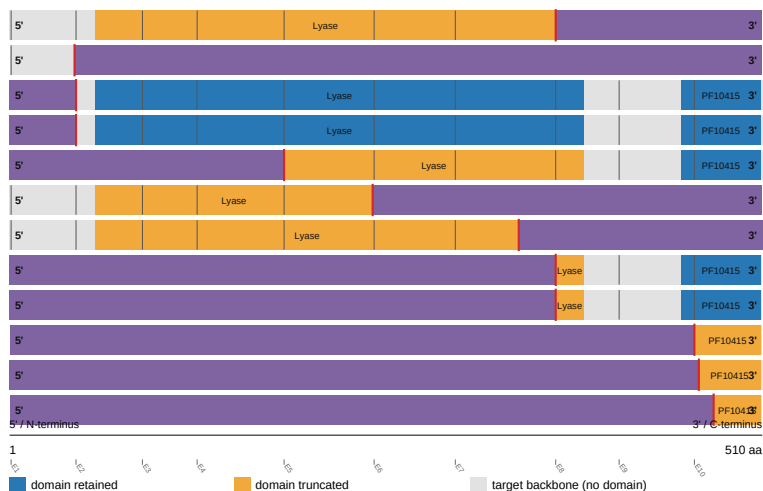
FH domain-retention track



## fusion schematic

FH fusion-transcript schematic

Fusion order varies by row: follow each row's 5' and 3' labels (FH portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

DGKH (x4) – Lyase truncated – Non-Small Cell Lung Cancer, Cancer of Unknown Primary, Ovarian Cancer

PDE1C – Lyase lost – Breast Cancer

RYY2 – Lyase retained – Non-Small Cell Lung Cancer

unknown – Lyase retained – Breast Cancer

NOTCH1 – Lyase truncated – Anal Cancer

RGS7 – Lyase truncated – Mesothelioma

FMN2 – Lyase truncated – Hepatobiliary C

ARID4B – Lyase truncated – Non-Small Cell Lur

RGS7 – Lyase truncated – Breast Cancer

DGKH – Lyase lost – Esophagogastric Ca

MAFTRR – Lyase lost – Colorectal Cancer

SLC35F3 – Lyase lost – Colorectal Cancer

breakpoint

## reference comparison

## Reference vs msk\_impact\_50k\_2026

